The Learning Premium in the Large-Horizon Limit

Propositions 1–3 (draft for *Orchestrating Agent Teams*)

Working draft

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Abstract

We show that, as the campaign horizon K grows with the budget per period $\rho = B/K$ held fixed, the per-period learning premium of the learn-then-exploit (ETC) team converges to an explicit function of the primitives: the value of the best binomial probing experiment, which coincides with the value of free perfect information $\mathrm{VPI}(\rho)$ outside a neighbourhood of the entry threshold. The limit does not depend on how the team schedules its learning, so the direction of the planner's reallocation in the limit is pinned down by primitives alone (Proposition 1 and Corollary 1). We then characterise the shape of the limit premium (Proposition 2): it is negative-to-zero at small budgets (information, even if free, lowers the value of a budget-paced team), positive and decaying like $\mathrm{Var}(\theta)/\rho^2$ at large budgets, and it opens with a kink, so the marginal-value ratio jumps at entry. The upper edge of the learning window is a finite-horizon effect. Finally (Proposition 3), when the decomposition and the full bandit (the Bayes-optimal policy that learns and executes in every auction) face the same ex-ante budget rule, both converge to the value of knowing the state from the start: the full bandit's total learning cost stays bounded, the decomposition's grows like $\log K$ (Proposition 4 shows this order cannot be improved by any probe-then-commit design), so the ratio of their learning premia tends to one and their allocations converge to the same limit. Under the paper's budget rule the decomposition keeps a gap that grows with the budget.

1 Setup and notation

We keep the model of the paper. The market state is $\theta \in \{\theta_L, \theta_H\}$ with $0 < \theta_L < \theta_H$ and prior $\pi_0 = \Pr(\theta = \theta_H) \in (0,1)$; $r = \theta_H/\theta_L$, $b^I = \sqrt{\theta_L\theta_H}$ and $\kappa_0 = \pi_0/(1-\pi_0)$. A bid b wins with probability $q_\theta(b) = b/(b+\theta)$ and pays b upon winning; $\bar{q}(\pi, b) = \pi q_{\theta_H}(b) + (1-\pi)q_{\theta_L}(b)$. A team has budget B and horizon K; we write $\rho = B/K$ for the budget per period and index values by K when the horizon matters.

Budget accounting and the rate function. As in the paper, whenever a team commits to an exploitation plan at belief π with remaining budget R over n periods, the plan spends R in expectation under π. By Lemma 2 of the paper (Exploitation phase) the optimal plan is a constant paced bid and its value is $n\, g(\pi; R/n)$, where for $\pi \in [0,1]$ and $\rho \geq 0$

$$g(\pi;\rho) := \bar{q}\big(\pi, b^*(\pi,\rho)\big), \qquad b^*(\pi,\rho)\,\bar{q}\big(\pi, b^*(\pi,\rho)\big) = \rho. \tag{1}$$

In particular the no-learning benchmark is $W_K(B) = K\, g(\pi_0; B/K)$.

Learn-then-commit protocols. A protocol in the class $\mathcal{L}$ probes at a fixed bid $b_p > 0$ for σ periods, where $\sigma \le K-1$ is a stopping time of the probe outcomes, and hands the number of wins w_σ to an exploitation agent that commits at the posterior π_σ to the paced plan with the remaining budget $R_\sigma = B - w_\sigma b_p \ge 0$ over the remaining $K-\sigma$ periods. By Lemma 2 of the paper its value is

$$V_K^\sigma(B) = \mathbb{E}\Big[w_\sigma + (K-\sigma)\, g\Big(\pi_\sigma; \frac{R_\sigma}{K-\sigma}\Big)\Big]. \tag{2}$$

The paper's ETC is the case $b_p = b^I$ and $\sigma \equiv \tau$ deterministic, with $V_K(B) = \max\{V_K^\tau(B) : 0 \le \tau \le \min(K-1, B/b^I)\}$; the adaptive-stopping version (draft DP) is $b_p = b^I$ with σ chosen optimally. Expectations are under the prior-predictive law, under which the posterior $(\pi_t)_t$ is a martingale with $\mathbb{E}[\pi_t] = \pi_0$.

Definition 1 (Limit objects)**.** For $\rho > 0$ let π_τ denote the posterior after τ probes at b^I and $\pi_\infty := \mathbf{1}_{\{\theta=\theta_H\}}$. Define

$$\begin{aligned}
\mathrm{VPI}(\rho) &:= \pi_0\, g(1;\rho) + (1-\pi_0)\, g(0;\rho) - g(\pi_0;\rho),\\
f_\tau(\rho) &:= \mathbb{E}\big[g(\pi_\tau;\rho)\big] - g(\pi_0;\rho) \quad (\tau \in \mathbb{N}), \qquad f_\infty := \mathrm{VPI},\\
\Psi(\rho) &:= \max\Big\{0, \sup_{\tau \in \mathbb{N}\cup\{\infty\}} f_\tau(\rho)\Big\},\\
\Phi(\rho) &:= \hat{g}(\pi_0;\rho) - g(\pi_0;\rho),
\end{aligned}$$

where $\hat{g}(\cdot;\rho)$ is the concave envelope of $g(\cdot;\rho)$ on $[0,1]$. Thus VPI is the per-period value of free perfect information, Ψ the value of the best binomial probing experiment at b^I (including none), and Φ the value of the best information structure. Since $\mathbb{E}[\pi_\tau] = \pi_0$ and $\hat{g}$ is concave, $\max\{\mathrm{VPI}(\rho), 0\} \le \Psi(\rho) \le \Phi(\rho)$.

Lemma 1 (Properties of g)**.** *(i) For every π, $\rho \mapsto g(\pi;\rho)$ is continuous, strictly increasing and strictly concave on $[0,\infty)$, with $g(\pi;0) = 0$ and $g < 1$.*

(ii) For every π and $S \ge 0$, $n \mapsto n\, g(\pi; S/n)$ is nondecreasing on $(0,\infty)$.

(iii) For $\rho > 0$, $g(\cdot;\rho) \in C^\infty([0,1])$, and g, $\partial_\pi g$, $\partial_\rho g$ are bounded on $[0,1] \times [\rho_a, \rho_b]$ for every $0 < \rho_a < \rho_b < \infty$.

Proof. (i) is shown in the proof of Lemma 2 of the paper. (ii) Concavity and $g(\pi;0) = 0$ imply that $x \mapsto g(\pi;x)/x$ is nonincreasing on $(0,\infty)$; hence $n\, g(\pi; S/n) = S \cdot g(\pi;S/n)/(S/n)$ is nondecreasing in n. (iii) $F(b,\pi,\rho) = b\,\bar{q}(\pi,b) - \rho$ is rational in (b,π) with positive denominators and $\partial_b F = \bar{q} + b\,\partial_b \bar{q} > 0$ for $b > 0$; the implicit function theorem gives a C^∞ solution $b^*(\pi,\rho)$ for $\rho > 0$, and boundedness follows from continuity on compact sets. □

Lemma 2 (Posterior concentration at b^I)**.** *Let $\beta := 2r^{1/4}/(1+\sqrt{r}) \in (0,1)$, the Bhattacharyya coefficient of the two outcome distributions at b^I. For every $\tau \in \mathbb{N}$, $\mathbb{E}\big[|\pi_\tau - \pi_\infty|\big] \le 2\sqrt{\pi_0(1-\pi_0)}\,\beta^\tau$.*

Proof. At b^I the win probabilities are $p_H = 1/(1+\sqrt{r})$ and $p_L = \sqrt{r}/(1+\sqrt{r})$, and the posterior odds are $\kappa_\tau = \kappa_0 r^{(\tau-2w)/2}$. Under θ_H, $1-\pi_\tau = 1/(1+\kappa_\tau) \le \kappa_\tau^{-1/2}$ and each probe multiplies $\kappa^{-1/2}$ by $r^{1/4}$ (win) or $r^{-1/4}$ (loss), so $\mathbb{E}[1-\pi_\tau \mid \theta_H] \le \kappa_0^{-1/2}\big(p_H r^{1/4} + (1-p_H) r^{-1/4}\big)^\tau = \kappa_0^{-1/2}\beta^\tau$. Symmetrically $\mathbb{E}[\pi_\tau \mid \theta_L] \le \kappa_0^{1/2}\beta^\tau$. Averaging with weights $\pi_0, 1-\pi_0$ gives the bound. Finally $\beta < 1$ because $1+\sqrt{r} > 2r^{1/4}$ for $r \ne 1$. □

2 Proposition 1: the large-horizon limit

Proposition 1 (Large-horizon limit of the learning premium)**.** *Fix $\rho > 0$ and let $B = \rho K$.*

(a) Achievability. *There are constants c_ρ, c'_ρ (depending on the primitives and ρ only) such that for every $\tau \in \mathbb{N}$ and all K large enough*

$$\frac{V_K^\tau(\rho K) - W_K(\rho K)}{K} \;\geq\; f_\tau(\rho) - c_\rho \frac{\tau}{K},$$

and for $\tau_K = \lceil \log K / \log(1/\beta) \rceil$,

$$\frac{V_K^{\tau_K}(\rho K) - W_K(\rho K)}{K} \;\geq\; \mathrm{VPI}(\rho) - c'_\rho \, \frac{1 + \log K}{K}.$$

Consequently $\liminf_K [V_K(\rho K) - W_K(\rho K)]/K \geq \Psi(\rho)$.

(b) Upper bound. *For every protocol in $\mathcal{L}$ and every K, B,*

$$\frac{V_K^\sigma(B) - W_K(B)}{K} \;\leq\; \frac{\mathbb{E}[\sigma]}{K} + \Phi(B/K),$$

and for the paper's ETC with a deterministic τ, $[V_K^\tau(B) - W_K(B)]/K \leq \tau/K + \Psi(B/K)$.

(c) Limit. *The optimal probing length satisfies $\tau_K^*/K \to 0$, and*

$$\lim_{K\to\infty} \frac{V_K(\rho K) - W_K(\rho K)}{K} = \Psi(\rho).$$

(d) Protocol invariance. *For every sequence of protocols in $\mathcal{L}$ with $\mathbb{E}[\sigma_K] = o(K)$,*

$$\limsup_K \; [V_K^{\sigma_K}(\rho K) - W_K(\rho K)]/K \;\leq\; \Phi(\rho).$$

On the set $\mathcal{R} := \{\rho : \Phi(\rho) = \max\{\mathrm{VPI}(\rho), 0\}\}$ the paper's ETC is asymptotically optimal in this class, and its limit premium $\max\{\mathrm{VPI}(\rho), 0\}$ depends on the primitives $(\theta_L, \theta_H, \pi_0)$ only.

Proof. *(a)* Fix τ; for K large, $\tau \leq K-1$ and $\tau b^I \leq \rho K$, so τ is feasible. Since $w \leq \tau$, the exploitation budget per period satisfies $R/(K-\tau) \geq \rho_K^\tau := \rho - \tau b^I/(K-\tau) \geq \rho/2$. Dropping the probe wins in (2) and using monotonicity and the Lipschitz bound $L_\rho := \sup\{\partial_\rho g(\pi; x) : \pi \in [0,1], x \in [\rho/2, \rho]\}$ from Lemma 1(iii),

$$V_K^\tau(\rho K) \geq (K-\tau)\,\mathbb{E}\big[g(\pi_\tau; \rho_K^\tau)\big] \geq (K-\tau)\,\mathbb{E}\big[g(\pi_\tau; \rho)\big] - L_\rho \tau b^I \geq K\,\mathbb{E}\big[g(\pi_\tau; \rho)\big] - \tau(1 + L_\rho b^I),$$

using $g < 1$. Subtracting $W_K(\rho K) = K g(\pi_0; \rho)$ gives the first bound with $c_\rho = 1 + L_\rho b^I$. For $\tau = \tau_K$, the mean value theorem with $M_\rho := \max_\pi |\partial_\pi g(\pi; \rho)|$ and Lemma 2 give $f_{\tau_K}(\rho) \geq \mathrm{VPI}(\rho) - 2M_\rho \sqrt{\pi_0(1-\pi_0)}\,\beta^{\tau_K} \geq \mathrm{VPI}(\rho) - M_\rho/K$, since $\beta^{\tau_K} \leq 1/K$; combining with the first bound and $\tau_K \leq 1 + \log K/\log(1/\beta)$ proves the second. Since $V_K \geq V_K^\tau$ for every feasible τ, $\liminf_K [V_K - W_K]/K \geq f_\tau(\rho)$ for all $\tau \in \mathbb{N} \cup \{\infty\}$, and $\tau = 0$ gives 0; hence the lim inf is at least $\Psi(\rho)$.

(b) In (2), $R_\sigma \leq B$, monotonicity and Lemma 1(ii) give $(K-\sigma)\, g(\pi_\sigma; R_\sigma/(K-\sigma)) \leq (K-\sigma)\, g(\pi_\sigma; B/(K-\sigma)) \leq K\, g(\pi_\sigma; B/K)$, and $w_\sigma \leq \sigma$. By optional stopping $\mathbb{E}[\pi_\sigma] = \pi_0$, so by concavity of $\hat{g}$, $\mathbb{E}[g(\pi_\sigma; \rho)] \leq \mathbb{E}[\hat{g}(\pi_\sigma; \rho)] \leq \hat{g}(\pi_0; \rho)$. Hence $V_K^\sigma(B) \leq \mathbb{E}[\sigma] + K\hat{g}(\pi_0; \rho)$; subtract $W_K(B) = K g(\pi_0; \rho)$. For deterministic τ at b^I, replace $\hat{g}(\pi_0; \rho)$ by $\mathbb{E}[g(\pi_\tau; \rho)] = g(\pi_0; \rho) + f_\tau(\rho) \leq g(\pi_0; \rho) + \Psi(\rho)$.

(c) Suppose $\limsup_K \tau_K^*/K > 0$ and pass to a subsequence with $\tau_K^*/K \to \varepsilon \in (0,1]$. Feasibility ($\tau b^I \le \rho K$) gives $\varepsilon b^I \le \rho$. Since $\tau_K^* \to \infty$, conditional on θ the strong law gives $w/\tau_K^* \to q_\theta(b^I)$ and Lemma 2 gives $\pi_{\tau_K^*} \to \pi_\infty$. With $s_\theta := b^I q_\theta(b^I)$, dominated convergence in (2) yields

$$\lim_K \frac{V_K^{\tau_K^*}(\rho K)}{K} = \mathbb{E}_\theta\Big[\varepsilon\, q_\theta(b^I) + (1-\varepsilon)\, g\Big(\pi_\infty; \tfrac{\rho - \varepsilon s_\theta}{1-\varepsilon}\Big)\Big]$$

(for $\varepsilon = 1$ the second term is absent). Because b^I is the paced bid for spend s_θ in a known state, $q_\theta(b^I) = g(\pi_\infty; s_\theta)$ on $\{\theta\}$, and strict concavity of $g(\pi_\infty;\cdot)$ gives $\varepsilon g(\pi_\infty; s_\theta) + (1-\varepsilon) g(\pi_\infty; \frac{\rho-\varepsilon s_\theta}{1-\varepsilon}) \le g(\pi_\infty;\rho)$ with equality only if $s_\theta = \rho$ (for $\varepsilon = 1$: $g(\pi_\infty; s_\theta) \le g(\pi_\infty;\rho)$ because $s_\theta < b^I \le \rho$). As $s_{\theta_L} \ne s_{\theta_H}$, the inequality is strict for at least one state, so $\lim_K V_K^{\tau_K^*}/K < \mathbb{E}_\theta[g(\pi_\infty;\rho)] = g(\pi_0;\rho) + \mathrm{VPI}(\rho) \le g(\pi_0;\rho) + \Psi(\rho)$, which by (a) is at most $\liminf_K V_K/K$ — contradicting $V_K = V_K^{\tau_K^*}$. Hence $\tau_K^* = o(K)$, and (b) gives $\limsup_K [V_K - W_K]/K \le \Psi(\rho)$; together with (a) the limit is $\Psi(\rho)$.

(d) The first claim is (b). On $\mathcal{R}$, $\max\{\mathrm{VPI}, 0\} \le \Psi \le \Phi = \max\{\mathrm{VPI}, 0\}$, so the ETC limit in (c) attains the upper bound. □

Corollary 1 (Allocation in the limit)**.** *Let $L(\rho) := g(\pi_0;\rho) + \Psi(\rho)$ for $\rho > 0$ and $L(0) = 0$. Fix $v_1 > v_2 > 0$ and $t > 0$, and let $T = tK$. Then $a_K^E/K = \alpha^E(t) := \arg\max_{\alpha\in[0,t]}\{v_1 g(\pi_0;\alpha) + v_2 g(\pi_0; t-\alpha)\}$ for every K, and every limit point of a_K^L/K maximizes $v_1 L(\alpha) + v_2 L(t-\alpha)$ over $\alpha \in [0,t]$. If that maximizer $\alpha^L(t)$ is unique, $(a_K^L - a_K^E)/K \to \alpha^L(t) - \alpha^E(t)$. At differentiability points and under the concavity condition of Theorem 1 of the paper, the sign of the limit shift is the sign of*

$$\frac{L'(\alpha^E)}{g_\rho(\pi_0;\alpha^E)} - \frac{L'(t-\alpha^E)}{g_\rho(\pi_0; t-\alpha^E)},$$

so the direction of the reallocation in the limit is determined by the primitives alone.

Proof. $W_K(\alpha K) = K g(\pi_0;\alpha)$ makes the benchmark problem independent of K. For each K, $\rho \mapsto V_K(\rho K)/K$ is nondecreasing (a larger budget enlarges the feasible set of τ and raises every exploitation value) and converges pointwise to L by Proposition 1(c). The family $\{f_\tau\}$ is equi-Lipschitz on compact subsets of $(0,\infty)$ (Lemma 1(iii)), so Ψ is continuous there, and $\Psi = 0$ near 0 by Proposition 2(a); thus L is continuous on $[0,\infty)$. Pointwise convergence of monotone functions to a continuous limit on a compact interval is uniform (Pólya), so the planner's normalised objectives converge uniformly on $[0,t]$ and limit points of maximisers maximise the limit objective. The sign statement is Theorem 1 of the paper applied to L, noting $W_K'(B) = g_\rho(\pi_0; B/K)$. □

Remark 1 (Learning while exploiting). Proposition 1 covers every learn-then-commit protocol. A team that updates after every auction and re-paces its remaining budget (certainty-equivalent pacing, "CE") is not in $\mathcal{L}$. In this model every bid is informative ($q_{\theta_L}(b) \ne q_{\theta_H}(b)$ for all $b > 0$), so such a team learns the state while exploiting, and the numerical evidence in Section 4 shows that its per-period premium converges to $\mathrm{VPI}(\rho)$ with a learning cost that grows much more slowly in K than that of fixed-τ ETC. A proof along the lines of (a) requires posterior concentration under bids bounded away from 0 and ∞ and control of the realised-wallet re-pacing; we leave it for future work.

3 Proposition 2: the shape of the limit premium

Let $m_k(\pi) := \pi\theta_H^{-k} + (1-\pi)\theta_L^{-k}$ and $\mu_k(\pi) := \pi\theta_H^k + (1-\pi)\theta_L^k$ (the k-th moments of $1/\theta$ and θ under belief π), and let $\mathrm{Var}_{\pi_0}(\theta) = \pi_0(1-\pi_0)(\theta_H - \theta_L)^2$.

Proposition 2 (Shape of the limit premium). *(a)* Small budgets. *As $\rho \downarrow 0$, uniformly in $\pi \in [0,1]$ together with the first two π-derivatives,*

$$g(\pi;\rho) = \sqrt{\rho\, m_1(\pi)} + O(\rho).$$

Hence $\mathrm{VPI}(\rho) = -c_1\sqrt{\rho} + O(\rho)$ *with* $c_1 := \sqrt{m_1(\pi_0)} - \pi_0\theta_H^{-1/2} - (1-\pi_0)\theta_L^{-1/2} > 0$, *and there is $\rho_c > 0$ such that $g(\cdot;\rho)$ is strictly concave on $[0,1]$ for all $\rho < \rho_c$; consequently $\Phi(\rho) = \Psi(\rho) = 0$ there: no information structure, free or costly, raises the value of a budget-paced team.*

(b) Large budgets. *As $\rho \to \infty$, uniformly in $\pi \in [0,1]$ together with the first two π-derivatives,*

$$g(\pi;\rho) = 1 - \frac{\mu_1(\pi)}{\rho} + \frac{\mu_1(\pi)^2 + \mu_2(\pi)}{\rho^2} + O(\rho^{-3}).$$

Hence $\mathrm{VPI}(\rho) = \mathrm{Var}_{\pi_0}(\theta)/\rho^2 + O(\rho^{-3})$, *and there is $\rho_v < \infty$ such that $g(\cdot;\rho)$ is strictly convex on $[0,1]$ for all $\rho > \rho_v$; consequently $\Psi(\rho) = \Phi(\rho) = \mathrm{VPI}(\rho) > 0$ there.*

(c) The learning window in the limit. *Let $\rho_1 := \inf\{\rho > 0 : \Psi(\rho) > 0\}$. Then $0 < \rho_c \le \rho_1 \le \rho_v < \infty$: learning has no value below ρ_1 and positive value at every $\rho > \rho_v$, decaying like $\mathrm{Var}_{\pi_0}(\theta)/\rho^2$. In the limit the window has no upper edge; in particular the finite-horizon upper edge $B_2(K)$ of the paper satisfies $B_2(K)/K \to \infty$.*

(d) Kink at entry. *If near ρ_1 one has $\Psi = \max\{0, f_\tau\}$ for some $\tau \in \mathbb{N} \cup \{\infty\}$ with $f_\tau(\rho_1) = 0 < f'_\tau(\rho_1)$, then L has a convex kink at ρ_1 and the limit marginal-value ratio $L'/g_\rho(\pi_0;\cdot)$ jumps from 1 to $1 + f'_\tau(\rho_1)/g_\rho(\pi_0;\rho_1)$.*

Proof. (a) Put $s = \sqrt{\rho}$ and $b = s\,x$. For $b < \theta_L$, $q_\theta(b) = \sum_{k\ge1}(-1)^{k-1}(b/\theta)^k$, so $b\,\bar q(\pi,b) = \sum_{k\ge1}(-1)^{k-1}m_k(\pi)\,b^{k+1}$ and the pacing equation $b\,\bar q = s^2$ reads

$$G(x,\pi,s) := \sum_{k\ge1}(-1)^{k-1}m_k(\pi)\,s^{k-1}x^{k+1} - 1 = 0.$$

G is real-analytic near $s = 0$, $G(x_0,\pi,0) = 0$ for $x_0(\pi) = m_1(\pi)^{-1/2}$, and $\partial_x G(x_0,\pi,0) = 2\sqrt{m_1(\pi)} > 0$. By the analytic implicit function theorem and compactness of $[0,1]$ there are $s_0 > 0$ and an analytic solution $x(\pi,s)$ on a neighbourhood of $[0,1]\times[0,s_0]$. Then $g(\pi;\rho) = \bar q(\pi, s\,x) = s\,h(\pi,s)$ with h analytic and $h(\pi,0) = m_1(\pi)x_0(\pi) = \sqrt{m_1(\pi)}$, which gives the expansion with all π-derivatives. Substituting $m_1(1) = \theta_H^{-1}$, $m_1(0) = \theta_L^{-1}$ yields the expansion of VPI; $c_1 > 0$ is Jensen's inequality for the strictly concave square root applied to the non-degenerate random variable $1/\theta$. Finally $\partial_\pi^2 g = -\sqrt{\rho}\,(m_1')^2/(4m_1^{3/2}) + O(\rho)$ with $m_1' = \theta_H^{-1} - \theta_L^{-1} \ne 0$ and $m_1 \in [\theta_H^{-1}, \theta_L^{-1}]$, so $g(\cdot;\rho)$ is strictly concave on $[0,1]$ for small ρ. Then $\hat g = g$, so $\Phi = 0$, and $f_\tau \le 0$ for every τ by Jensen, so $\Psi = 0$.

(b) Put $u = 1/\rho$ and $b = y/u$. Since $q_\theta(b) = 1 - u\theta/(y+u\theta)$, the pacing equation reads $H(y,\pi,u) := y - y\,\mathbb{E}_\pi[u\theta/(y+u\theta)] - 1 = 0$, where $\mathbb{E}_\pi$ averages over θ with $\Pr(\theta_H) = \pi$. H is analytic near $u = 0$, $H(1,\pi,0) = 0$ and $\partial_y H(1,\pi,0) = 1$, so $y(\pi,u)$ is analytic near $[0,1]\times\{0\}$ and so is $g = 1 - \mathbb{E}_\pi[u\theta/(y+u\theta)]$. Expanding, $y = 1 + u\mu_1 - u^2\mu_2 + O(u^3)$ and $g = 1 - u\mu_1 + u^2(\mu_1^2 + \mu_2) + O(u^3)$. For a known state ($\pi \in \{0,1\}$) this is $1 - u\theta + 2u^2\theta^2 + O(u^3)$, and averaging over the prior, $\mathrm{VPI} = u^2\big(2\mu_2 - \mu_1^2 - \mu_2\big) + O(u^3) = u^2\,\mathrm{Var}_{\pi_0}(\theta) + O(u^3)$, where $\mu_k = \mu_k(\pi_0)$. Since μ_1, μ_2 are affine in π, $\partial_\pi^2 g = 2u^2(\theta_H - \theta_L)^2 + O(u^3) > 0$ for small u, so $g(\cdot;\rho)$ is strictly convex. Then $\hat g$ is the chord, so $\Phi = \mathrm{VPI}$; and because $\pi_{\tau+1}$ is a mean-preserving spread of π_τ, $\mathbb{E}[g(\pi_\tau;\rho)]$ is nondecreasing in τ with limit $\pi_0 g(1;\rho) + (1-\pi_0)g(0;\rho)$, so $\sup_\tau f_\tau = \mathrm{VPI}$ and $\Psi = \mathrm{VPI}$. Strict convexity also gives $\mathrm{VPI} > 0$.

(c) By (a), $\Psi = 0$ on $(0, \rho_c)$, so $\rho_1 \geq \rho_c > 0$; by (b), $\Psi > 0$ on (ρ_v, ∞), so $\rho_1 \leq \rho_v$. For any fixed $\rho > \rho_v$, Proposition 1(a) gives $[V_K(\rho K) - W_K(\rho K)]/K \geq \mathrm{VPI}(\rho) - c'_\rho(1 + \log K)/K > 0$ for K large, so the ETC team learns at budget ρK and $B_2(K) \geq \rho K$ eventually; as ρ is arbitrary, $B_2(K)/K \to \infty$.

(d) Near ρ_1, $L = g(\pi_0; \cdot) + \max\{0, f_\tau\}$. Its left derivative at ρ_1 is $g_\rho(\pi_0; \rho_1)$ and its right derivative is $g_\rho(\pi_0; \rho_1) + f'_\tau(\rho_1)$, a convex kink. Dividing by $g_\rho(\pi_0; \rho_1)$ gives the jump of the ratio. □

Remark 2 (Interpretation). Part (a) is the limit form of the cross-state subsidy: at small budgets the uninformed paced bid spends more in the cheap state, where the marginal dollar buys more impressions, and any information that splits the team's belief removes this subsidy because each posterior must balance its own budget. Under an ex-ante budget constraint, by contrast, free information is never harmful (Blackwell). Part (b) is decision insensitivity: at large budgets both states are won almost surely and the value of information decays like $\mathrm{Var}(\theta)/\rho^2$, but it never vanishes; the upper edge of the paper's learning window comes from the cost $O(\log K)$ of probing at b^I, which is spread over the horizon and vanishes per period as $K \to \infty$. Together with Corollary 1, the two allocation regimes of the paper follow in the limit: if the junior team's budget per period is below ρ_1 and the senior's is where the limit ratio exceeds one, learning skews the split toward the senior team; once both are above ρ_1 and the limit ratio is decreasing between them, learning balances the split.

4 Numerical verification

All computations use the paper's baseline $\theta_L = \frac{1}{2}$, $\theta_H = 8$ ($r = 16$, $b^I = 2$), $\pi_0 = \frac{1}{2}$ and $v_1/v_2 = 2$; code is in `bandit_check/limit_theory.py` and `bandit_check/run_large_K.py`, which writes every number reported below to `bandit_check/results/large_K/summary.txt`. The finite-K values are the paper's exact ETC (integer τ, binomial w; $\tau \leq 150$ for $K = 400$) and, for comparison, a learn-while-doing team that updates after every auction and re-paces its realised budget ("CE", computed by dynamic programming over periods, budget and posterior log-odds).

Limit objects and thresholds. The rate function is strictly concave in the belief only for $\rho < 0.035$ ($= \rho_c$ on the grid) and strictly convex in the belief for every $\rho \geq 1.375$ ($= \rho_v$ on the grid; checked up to $\rho = 10^4$); the best information structure starts to pay at $\rho = 0.305$ ($\Phi > 0$); both the best binomial experiment and perfect information start to pay at $\rho_1 = 0.4558$. Hence $\rho_c = 0.035 \leq \rho_1 = 0.4558 \leq \rho_v = 1.375$, as in Proposition 2(c). The entering branch is $\tau = \infty$, and on the grid $\Psi = \max\{\mathrm{VPI}, 0\}$ for every ρ: *the limit of the paper's ETC is the value of free perfect information itself.* Φ exceeds $\max\{\mathrm{VPI}, 0\}$ only on $(0.305, 0.8)$, by at most 0.0017 impressions per period (at $\rho \approx 0.455$). The asymptotics of Proposition 2 are confirmed: $\mathrm{VPI}(\rho)/\sqrt{\rho} = -0.1428$ at $\rho = 10^{-4}$ (theory: $-c_1 = -0.1469$) and $\rho^2 \mathrm{VPI}(\rho) = 13.89$ at $\rho = 3000$ (theory: $\mathrm{Var}(\theta) = 14.06$). At entry, $\mathrm{VPI}'(\rho_1) = 0.0378$ and $g_\rho(\pi_0; \rho_1) = 0.2526$, so the limit marginal-value ratio jumps from 1 to 1.150 (the finite-K counterparts are 1.22 for the smoothed ETC and 1.17 for CE at $K = 100$).

Rates. The total learning cost $K\,[\Psi(\rho) - (V_K - W_K)/K]$ of the ETC team grows like $\log K$, as in Proposition 1(a): at $\rho = 2.5$ it is 0.50, 0.65, 0.80, 0.95 impressions for $K = 50, 100, 200, 400$, and 0.75, 1.19, 1.68, 2.20 at $\rho = 10$. The CE team's cost (relative to VPI) is much smaller and grows more slowly: 0.28, 0.32, 0.35, 0.39 at $\rho = 2.5$ and 0.14, 0.16, 0.18, 0.20 at $\rho = 10$. The optimal probing length grows roughly like $\log K$, so $\tau^*_K/K \to 0$ as in Proposition 1(c): at $\rho = 1.5$, $\tau^*_K = 7, 10, 13, 15$ and $\tau^*_K/K = 0.14, 0.10, 0.065, 0.037$. Near entry, the adaptive-stopping

Table 1: Per-period learning premium $(V_K - W_K)/K$ against the limit.

			ETC (exact), $K =$				CE (always learning), $K =$			
ρ	VPI	Ψ	50	100	200	400	50	100	200	400
0.20	−0.0113	0	0	0	0	0	−0.0224	−0.0175	−0.0147	−0.0133
0.60	0.0049	0.0049	0	0.0010	0.0021	0.0030	−0.0041	−0.0001	0.0022	0.0034
1.00	0.0143	0.0143	0.0090	0.0110	0.0123	0.0132	0.0066	0.0100	0.0119	0.0130
2.50	0.0252	0.0252	0.0151	0.0187	0.0212	0.0228	0.0195	0.0220	0.0234	0.0242
6.00	0.0228	0.0228	0.0068	0.0114	0.0153	0.0181	0.0190	0.0207	0.0216	0.0222
10.0	0.0174	0.0174	0.0025	0.0055	0.0090	0.0119	0.0146	0.0158	0.0165	0.0169

ETC exceeds Ψ and moves toward Φ, as allowed by Proposition 1(d): at $\rho = 0.5$ its premium is 0, 0.0011, 0.0019, 0.0023 against $\Psi = 0.0016$ and $\Phi = 0.0026$.

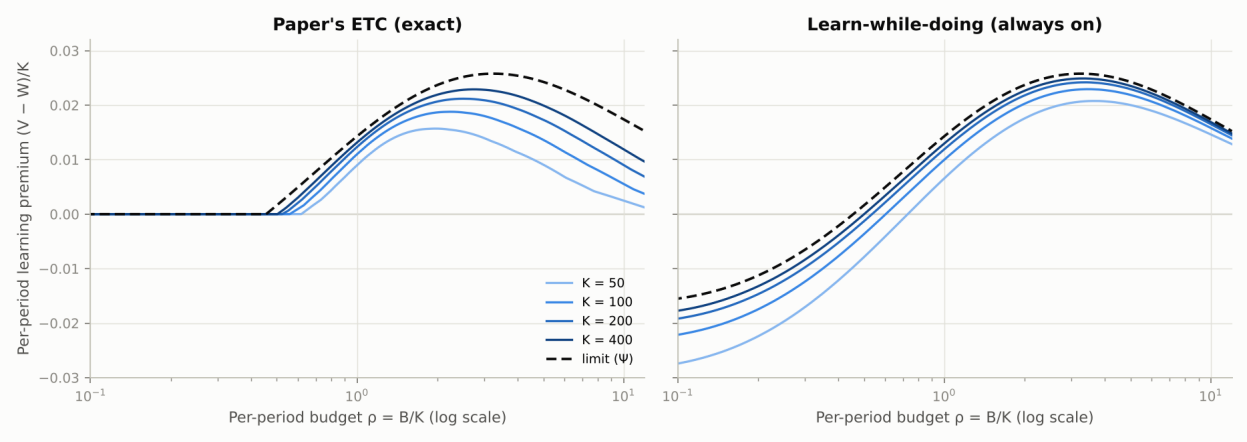


Figure 1: Per-period learning premium for $K = 50, 100, 200, 400$ (light to dark) and the limit (dashed). Left: the paper's ETC converges to $\Psi = \max\{\text{VPI}, 0\}$. Right: a learn-while-doing team converges to VPI, which is negative below ρ_1 (free information lowers the value of a budget-paced team).

Allocation. Table 2 reports the total budget per period $t = T/K$ at which learning first moves the split (toward the senior team) and at which the shift turns negative (toward the junior team). Both switch points converge to the same limit for the two protocols, as Corollary 1 predicts; at the limit switch $t = 1.5$ the junior team's benchmark budget per period is ≈ 0.48, just above ρ_1. The CE team's normalised shift converges quickly (e.g. $-0.0329, -0.0319, -0.0315, -0.0314$ at $t = 2.5$, limit -0.0311); the ETC team's converges more slowly, and its finite-K "re-concentration" at large t moves to ever larger budgets (Figure 2), consistent with Proposition 2(c).

Open points. (i) Uniqueness of the zero of VPI and transversality at entry are verified numerically here and across the parameter sweep, but not proved. (ii) A proof of the CE limit. (iii) A probing design that implements the concave envelope (e.g. a planner-chosen probe bid with belief-dependent stopping) would lower the limit entry point from $\rho_1 = 0.456$ toward 0.305; Proposition 1(b) shows that no learn-then-commit protocol can do better than Φ.

Table 2: Switch points of the allocation shift $(a_K^L - a_K^E)/K$ in $t = T/K$ ($v_1/v_2 = 2$).

	$K = 50$	$K = 100$	$K = 200$	$K = 400$	limit
ETC: shift becomes positive	0.90	0.85	0.80	0.75	0.65
ETC: shift turns negative	1.95	1.85–1.95	1.75	1.65–1.70	1.50
CE: shift becomes positive	1.05	0.90	0.80	0.75	0.65
CE: shift turns negative	2.30	1.95	1.75	1.65	1.50

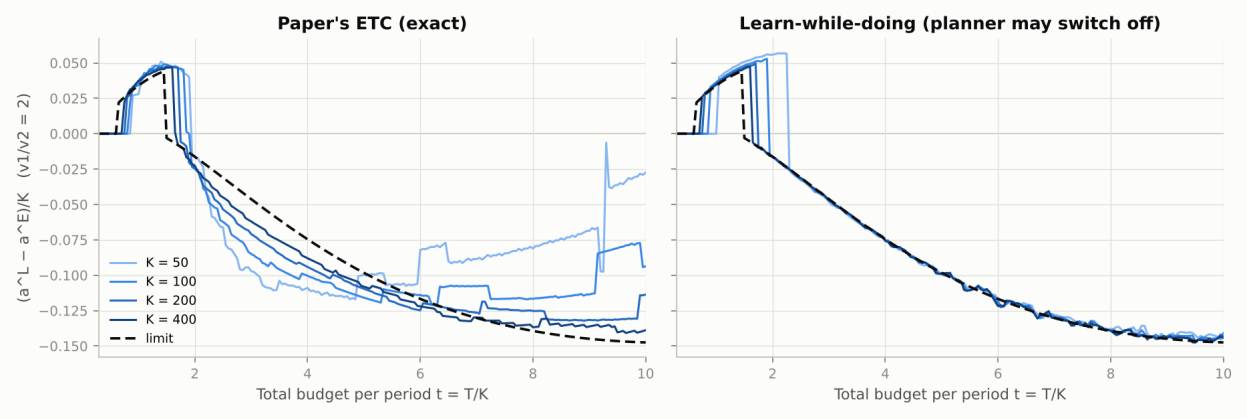


Figure 2: Normalised allocation shift $(a_K^L - a_K^E)/K$ for $K = 50, 100, 200, 400$ and the limit of Corollary 1 (dashed). Both protocols converge to the same limit.

5 Proposition 3: the decomposition and the full bandit under a common budget rule

The full bandit, the Bayes-optimal policy that learns and executes in every auction, is computed by dynamic programming under an *ex-ante* budget rule: the expected total spend may not exceed B. To compare it with the decomposition we hold the budget rule fixed and give the decomposition the same rule.

Ex-ante budget rule. A bidding policy chooses $b_t \geq 0$ in auction $t = 1, \ldots, K$ as a (possibly randomised) function of past bids and outcomes. With $c_t = b_t \mathbf{1}_{\{\text{win}_t\}}$ the payment in auction t, the policy is feasible for budget B if $\mathbb{E}[\sum_t c_t] \leq B$ under the prior-predictive law. Let

$$V_K^{\text{FB}}(B) := \sup\Big\{\mathbb{E}\Big[\sum_t \mathbf{1}_{\{\text{win}_t\}}\Big] : \text{ feasible policies}\Big\},$$

the *full bandit*, and let $V_K^{\text{D}}(B)$ be the same supremum over the *decomposition* class $\mathcal{D}$: bid b^I for τ auctions, then a constant bid $b(w)$ that depends only on the number of probe wins w, for $\tau \in \{0, \ldots, K-1\}$ and any function $b(\cdot) \geq 0$. The benchmark is unchanged: a policy that does not use outcomes is best off with a constant paced bid, so it earns $W_K(B) = K\, g(\pi_0; B/K)$.

The limit object. Write $g_H(s) := g(1; s)$ and $g_L(s) := g(0; s)$ for the per-period win rate at per-period spend s when the state is known, and $\pi_H = \pi_0$, $\pi_L = 1 - \pi_0$. Define

$$\mathcal{L}(\rho) := \max\Big\{\pi_H g_H(s_H) + \pi_L g_L(s_L) : \; s_H, s_L \geq 0, \; \pi_H s_H + \pi_L s_L \leq \rho\Big\}, \qquad \Lambda(\rho) := \mathcal{L}(\rho) - g(\pi_0; \rho),$$

the per-period value of knowing the state from the start under the ex-ante rule, and its excess over the benchmark. The maximum exists (compact feasible set, continuous objective); $\mathcal{L}$ is continuous, increasing and strictly concave, with $\mathcal{L}(0) = 0$ (if $\rho_1 \neq \rho_2$ have maximisers $s^1 \neq s^2$, their midpoint is feasible for $(\rho_1 + \rho_2)/2$ and strictly better than the average, by strict concavity of g_H, g_L).

Lemma 3 (Posterior concentration under adaptive bids)**.** *Let*

$$\beta(b) := \sqrt{q_H(b)\, q_L(b)} + \sqrt{(1 - q_H(b))(1 - q_L(b))},$$

which is < 1 for every $b > 0$, and let the bids $b_1, b_2, \ldots$ be predictable with values in a set on which $\beta \leq \bar{\beta} < 1$. Then

$$\mathbb{E}[1 - \pi_t \mid \theta_H] \leq \kappa_0^{-1/2} \bar{\beta}^{\,t}, \qquad \mathbb{E}[\pi_t \mid \theta_L] \leq \kappa_0^{1/2} \bar{\beta}^{\,t},$$

and the probability that the more likely state under π_t is not the true one is at most $4\sqrt{\pi_0(1-\pi_0)}\, \bar{\beta}^{\,t}$.

Proof. Under θ_H, $Z_t := \kappa_t^{-1/2}$ (with κ_t the posterior odds of θ_H) satisfies

$$\mathbb{E}[Z_{t+1} \mid \mathcal{F}_t] = Z_t \sum_y p_H(y) \sqrt{p_L(y)/p_H(y)} = Z_t\, \beta(b_{t+1}) \leq \bar{\beta}\, Z_t,$$

where y ranges over win and loss at the bid b_{t+1}, which is $\mathcal{F}_t$-measurable. Hence $\mathbb{E}[Z_t] \leq \kappa_0^{-1/2} \bar{\beta}^{\,t}$, and $1 - \pi_t \leq Z_t$ as in Lemma 2; the case θ_L is symmetric. Misclassification under θ_H means $\pi_t < 1/2$, whose probability is at most $2\mathbb{E}[1 - \pi_t \mid \theta_H]$ by Markov's inequality; averaging the two states gives the last bound. $\beta(b) < 1$ because $q_H(b) \neq q_L(b)$ for $b > 0$. □

Proposition 3 (Convergence under a common budget rule)**.** *Fix $\rho > 0$ and let $B = \rho K$. Under the ex-ante budget rule:*

(a) Ordering. $W_K(\rho K) \leq V_K^{\mathrm{D}}(\rho K) \leq V_K^{\mathrm{FB}}(\rho K) \leq K\mathcal{L}(\rho)$ *for every K.*

(b) Rates. *There are constants C_ρ, C'_ρ such that for all K large enough*

$$K\mathcal{L}(\rho) - V_K^{\mathrm{FB}}(\rho K) \leq C'_\rho, \qquad K\mathcal{L}(\rho) - V_K^{\mathrm{D}}(\rho K) \leq C_\rho\,(1 + \log K).$$

The full bandit's total learning cost is bounded; the decomposition's grows at most logarithmically.

(c) Learning premia. $\Lambda(\rho) > 0$, $[V_K^X(\rho K) - W_K(\rho K)]/K \to \Lambda(\rho)$ *for $X \in \{\mathrm{D}, \mathrm{FB}\}$, and*

$$1 \;\geq\; \frac{V_K^{\mathrm{D}}(\rho K) - W_K(\rho K)}{V_K^{\mathrm{FB}}(\rho K) - W_K(\rho K)} \;\geq\; 1 - \frac{C_\rho(1 + \log K)}{K\,\Lambda(\rho)} \;\to\; 1,$$

uniformly on compact subsets of $(0, \infty)$.

(d) Where convergence is slow. $\Lambda(\rho) = O(\rho^{3/2})$ *as $\rho \downarrow 0$, and $\Lambda(\rho) = \mathrm{Var}_{\pi_0}(\sqrt{\theta})/\rho + O(\rho^{-2})$ as $\rho \to \infty$. For fixed K the bound in (c) is therefore weakest at small and at large budgets, and the set of budgets on which the ratio exceeds any level $1 - \delta$ widens as K grows.*

(e) Allocation. *Fix $v_1 > v_2 > 0$, $t > 0$ and $T = tK$. For $X \in \{\mathrm{D}, \mathrm{FB}\}$ the planner's allocation satisfies $a_K^X/K \to \alpha^{\mathcal{L}}(t)$, the unique maximiser of $v_1\mathcal{L}(\alpha) + v_2\mathcal{L}(t - \alpha)$ on $[0, t]$, while $a_K^E/K = \alpha^E(t)$ for every K. Hence $(a_K^{\mathrm{D}} - a_K^{\mathrm{FB}})/K \to 0$, and whenever $\alpha^{\mathcal{L}}(t) \neq \alpha^E(t)$ the two learning teams shift the split in the same direction for all K large enough; that direction is the sign of $\mathcal{L}'(\alpha^E)/g_\rho(\pi_0; \alpha^E) - \mathcal{L}'(t - \alpha^E)/g_\rho(\pi_0; t - \alpha^E)$.*

Proof. *(a)* The policy with $\tau = 0$ and the constant paced bid $b^*(\pi_0, \rho)$ lies in $\mathcal{D}$, spends exactly ρK in expectation and earns W_K; and $\mathcal{D}$ is a subset of all policies. For the last inequality fix any feasible policy and condition on θ. In auction t the bid b_t is chosen before the outcome, so $\mathbb{E}[\mathbf{1}_{\{\text{win}_t\}} \mid \theta, \mathcal{F}_{t-1}, b_t] = q_\theta(b_t) = g_\theta(s_t)$ with $s_t := b_t q_\theta(b_t)$, because in a known state the bid that spends s_t per period is b_t itself. By Jensen's inequality for the concave g_θ, the expected wins given θ are at most $K g_\theta(S_\theta/K)$, where $S_\theta = \mathbb{E}[\sum_t c_t \mid \theta]$. Feasibility is $\pi_H S_H + \pi_L S_L \le \rho K$, so the policy earns at most $K\mathcal{L}(\rho)$.

(b) Let (s_H^*, s_L^*) attain $\mathcal{L}(\rho)$; the budget binds, $\pi_H s_H^* + \pi_L s_L^* = \rho$, and $s_H^*, s_L^* > 0$ because $g_\theta'(0+) = \infty$. For $\varepsilon \in (0,1)$ let b_θ be the bid that spends $(1-\varepsilon)s_\theta^*$ per period in the known state θ, and $\bar{b} := \max_\theta b^*(\theta; s_\theta^*) \ge b_\theta$. Two policies use these bids.

Full bandit. In every auction bid $b_{\hat{\theta}_t}$, where $\hat{\theta}_t$ is the more likely state under π_{t-1}. For $\varepsilon \in [0, \frac{1}{2}]$ the two bids lie in a compact subset of $(0,\infty)$ that does not depend on ε, so Lemma 3 applies with one $\bar{\beta} < 1$ for all such ε, and the misclassification probabilities e_t satisfy $\sum_t e_t \le E := 2/(1-\bar{\beta})$. When $\hat{\theta}_t = \theta$ the auction spends $(1-\varepsilon)s_\theta^*$ and wins with probability $g_\theta((1-\varepsilon)s_\theta^*)$; otherwise it spends at most $\bar{b}$ and wins with probability ≥ 0. The expected spend is at most $K(1-\varepsilon)\rho + \bar{b}E$, which is $\le K\rho$ for $\varepsilon = \bar{b}E/(\rho K)$, which is at most $\frac{1}{2}$ for K large. Using $g_\theta((1-\varepsilon)s) \ge (1-\varepsilon)g_\theta(s)$ (concavity and $g_\theta(0) = 0$) and $g_\theta \le 1$, the expected wins are at least $\sum_t[(1-\varepsilon)\mathcal{L}(\rho) - e_t] \ge K\mathcal{L}(\rho) - \varepsilon K - E$, so $K\mathcal{L}(\rho) - V_K^{\mathrm{FB}} \le C'_\rho := E(1 + \bar{b}/\rho)$.

Decomposition. Probe at b^I for $\tau_K = \lceil \log K / \log(1/\beta) \rceil$ auctions, with $\beta = \beta(b^I)$ as in Lemma 2; then bid $b_{\hat{\theta}}$, $\hat{\theta}$ the more likely state after the probes (a function of w), for the remaining $K - \tau_K$ auctions. By Lemma 3 the misclassification probability is $e \le 2\beta^{\tau_K} \le 2/K$. The probes spend at most $\tau_K b^I$, so the expected spend is at most $\tau_K b^I + K(1-\varepsilon)\rho + K\bar{b}e$, which is $\le K\rho$ for $\varepsilon = (\tau_K b^I + 2\bar{b})/(\rho K)$. The expected wins are at least $(K - \tau_K)[(1-\varepsilon)\mathcal{L}(\rho) - e] \ge K\mathcal{L}(\rho) - \tau_K - \varepsilon K - 2$, and $\tau_K \le 1 + \log K/\log(1/\beta)$ gives $K\mathcal{L}(\rho) - V_K^{\mathrm{D}} \le (1 + b^I/\rho)(1 + \log K/\log(1/\beta)) + 2\bar{b}/\rho + 2$, which is at most $C_\rho(1 + \log K)$ with $C_\rho = (1 + b^I/\rho)(1 + 1/\log(1/\beta)) + 2 + 2\bar{b}/\rho$.

(c) Divide (a)–(b) by K and subtract $W_K/K = g(\pi_0; \rho)$. For the ratio, the numerator is at least $K\Lambda - C_\rho(1 + \log K)$ and the denominator at most $K\Lambda$. $\Lambda(\rho) > 0$: the uninformed bid b_0 is a feasible point of the problem defining $\mathcal{L}$ (it spends $s_\theta = b_0 q_\theta(b_0)$ in state θ and earns $g(\pi_0; \rho)$), but it is not optimal, because the marginal value of spend in a known state at bid b,

$$g'_\theta(s) = \frac{q'_\theta(b)}{q_\theta(b) + b\, q'_\theta(b)} = \frac{\theta}{b(b + 2\theta)},$$

is strictly increasing in θ at the common bid b_0, so moving spend towards θ_H raises the value. Uniformity: C_ρ is bounded and Λ bounded away from zero on compact subsets of $(0,\infty)$.

(d) Small budgets. Let $\mu_0 = g_\rho(\pi_0; \rho)$ and $\phi_\theta(b) := q_\theta(b)(1 - \mu_0 b)$. The uninformed bid b_0 maximises $\pi_H \phi_H + \pi_L \phi_L$ and spends ρ, while by weak duality $\mathcal{L}(\rho) \le \pi_H \max_b \phi_H + \pi_L \max_b \phi_L + \mu_0 \rho$. Hence $\Lambda(\rho) \le \sum_\theta \pi_\theta[\phi_\theta(b_\theta^\star) - \phi_\theta(b_0)]$ with $b_\theta^\star = -\theta + \sqrt{\theta^2 + \theta/\mu_0} = \frac{1}{2\mu_0} - \frac{1}{8\theta\mu_0^2} + O(\mu_0^{-3})$. Both $b_\theta^\star$ and b_0 lie in an interval of length $O(\mu_0^{-2})$, on which $|\phi''_\theta| = O(\mu_0)$, so $\Lambda(\rho) = O(\mu_0 \cdot \mu_0^{-4}) = O(\mu_0^{-3})$, and $\mu_0 \sim \frac{1}{2}\sqrt{m_1(\pi_0)/\rho}$ by Proposition 2(a). *Large budgets.* In a known state $b\, q_\theta(b) = s$ gives $b = \frac{1}{2}(s + \sqrt{s^2 + 4s\theta}) \in [s, s + \theta]$, hence $1 - \frac{\theta}{s} \le 1 - \frac{\theta}{s+\theta} \le g_\theta(s) \le 1 - \frac{\theta}{s+2\theta}$. The lower bound at $s_\theta = \rho\sqrt{\theta}/\mathbb{E}\sqrt{\theta}$ gives $\mathcal{L}(\rho) \ge 1 - (\mathbb{E}\sqrt{\theta})^2/\rho$; the upper bound and the Cauchy–Schwarz inequality $\sum_\theta \pi_\theta \theta / x_\theta \ge (\mathbb{E}\sqrt{\theta})^2 / \sum_\theta \pi_\theta x_\theta$ give $\mathcal{L}(\rho) \le 1 - (\mathbb{E}\sqrt{\theta})^2/(\rho + 2\mu_1)$. With $g(\pi_0; \rho) = 1 - \mu_1/\rho + O(\rho^{-2})$ from Proposition 2(b), $\Lambda(\rho) = [\mu_1 - (\mathbb{E}\sqrt{\theta})^2]/\rho + O(\rho^{-2}) = \mathrm{Var}(\sqrt{\theta})/\rho + O(\rho^{-2})$.

(e) As in Corollary 1: $\alpha \mapsto V_K^X(\alpha K)/K$ is nondecreasing and converges pointwise to the continuous $\mathcal{L}$ by (c), hence uniformly on $[0, t]$ (Pólya); limit points of a_K^X/K maximise the limit objective, whose maximiser is unique because $\mathcal{L}$ is strictly concave. The sign statement is the

first-order condition at α^E, where $v_1 g_\rho(\pi_0;\alpha^E) = v_2 g_\rho(\pi_0; t-\alpha^E)$, combined with strict concavity; $\mathcal{L}$ is differentiable with $\mathcal{L}'(\rho)$ equal to the multiplier of its budget constraint. □

Proposition 4 (The decomposition's learning cost is of order $\log K$)**.** *Under the ex-ante budget rule fix $\rho > 0$ and let $\mu^* := \mathcal{L}'(\rho) > 0$. For $\theta \in \{H, L\}$ let $\phi_\theta(b) := q_\theta(b)(1-\mu^* b)$, $\ell_\theta := \max_{b\geq 0} \phi_\theta(b)$ and $r_\theta(b) := \ell_\theta - \phi_\theta(b) \geq 0$, and for a bid $b > 0$ let $\mathrm{kl}_\theta(b)$ be the Kullback–Leibler divergence of the win indicator at b under θ from that under the other state. Consider any feasible policy that bids a fixed $b_p > 0$ until a stopping time $\sigma \leq K$ of these probe outcomes and then follows a bidding plan, possibly time-varying and randomised, that uses the probe outcomes but no later ones. Then*

$$c_\rho := \max_\theta \frac{\pi_\theta\, r_\theta(b_p)}{2\,\mathrm{kl}_\theta(b_p)} > 0 \qquad \text{and} \qquad K\mathcal{L}(\rho) - V_K(\rho K) \;\geq\; c_\rho \log K \quad \text{for all } K \geq K_\rho.$$

This covers the decomposition ($b_p = b^I$, deterministic τ), its adaptive-stopping version, and the same structure with any other fixed probe bid; c_ρ and K_ρ do not depend on the policy beyond b_p. With Proposition 3(b),

$$c_\rho \log K \;\leq\; K\mathcal{L}(\rho) - V_K^{\mathrm{D}}(\rho K) \;\leq\; C_\rho(1+\log K), \qquad V_K^{\mathrm{FB}}(\rho K) - V_K^{\mathrm{D}}(\rho K) \;\geq\; c_\rho \log K - C'_\rho \;\to\; \infty.$$

Proof. Step 1: Lagrangian regret. For a feasible policy, $\mathbb{E}[\text{wins}] \leq \mathbb{E}[\text{wins}] - \mu^*(\mathbb{E}[\text{spend}] - \rho K) = \mu^*\rho K + \mathbb{E}\sum_t \phi_\theta(b_t)$, because given θ and b_t auction t yields expected wins minus μ^* times expected payment equal to $\phi_\theta(b_t)$. The problem defining $\mathcal{L}$ is concave with a Slater point, so strong duality holds with multiplier μ^*; substituting $s = b\,q_\theta(b)$, $\max_s[g_\theta(s) - \mu^* s] = \ell_\theta$ and $\mathcal{L}(\rho) = \mu^*\rho + \pi_H\ell_H + \pi_L\ell_L$. Hence $K\mathcal{L}(\rho) - V \geq \mathrm{Reg} := \mathbb{E}\sum_t r_\theta(b_t)$, and $\mathrm{Reg} = \pi_H \mathrm{Reg}_H + \pi_L \mathrm{Reg}_L$ with $\mathrm{Reg}_\theta := \mathbb{E}_\theta \sum_t r_\theta(b_t)$. Both sides are linear in a randomisation that is independent of θ, so it suffices to treat deterministic plans.

Step 2: no bid is optimal in both states. Writing $\phi_\theta(b) = (1+\mu^*\theta) - \mu^* b - \theta(1+\mu^*\theta)/(b+\theta)$ shows that ϕ_θ is strictly concave with unique maximiser $b^\star_\theta = \sqrt{\theta^2 + \theta/\mu^*} - \theta = \mu^{*-1}/(1+\sqrt{1+1/(\theta\mu^*)})$, which is strictly increasing in θ. Hence $r_H(b) + r_L(b) > 0$ for every $b \geq 0$; it is continuous and tends to ∞ as $b \to \infty$, so $D := \min_{b\geq 0}[r_H(b) + r_L(b)] > 0$. For the same reason $r_\theta(b_p) > 0$ for at least one θ, so $c_\rho > 0$.

Step 3: a test. Let θ' attain c_ρ, θ'' be the other state, and $n := \mathbb{E}_{\theta'}[\sigma]$. Every probe costs $r_{\theta'}(b_p)$ in state θ', so $\mathrm{Reg}_{\theta'} \geq r_{\theta'}(b_p)\, n$. After σ, each bid satisfies $r_{\theta'}(b_t) + r_{\theta''}(b_t) \geq D$, so at least one term is $\geq D/2$; let N'' be the number of auctions after σ with $r_{\theta''}(b_t) \geq D/2$ (a function of the probe outcomes). Declare θ' if $\sigma \leq K/2$ and $N'' \geq (K-\sigma)/2$, and θ'' otherwise; the decision is measurable with respect to the probe outcomes up to the stopping time $\sigma' := \sigma \wedge \lfloor K/2 \rfloor$. Under θ'', declaring θ' forces at least $(K-\sigma)/2 \geq K/4$ auctions with regret $\geq D/2$, so $P_{\theta''}(\text{declare } \theta') \leq 8\mathrm{Reg}_{\theta''}/(DK)$. Under θ', declaring θ'' means either $\sigma > K/2$, which has probability at most $2n/K$, or $N'' < (K-\sigma)/2$ with $\sigma \leq K/2$, which forces at least $K/4$ auctions with $r_{\theta'} \geq D/2$; hence $P_{\theta'}(\text{declare } \theta'') \leq 2n/K + 8\mathrm{Reg}_{\theta'}/(DK)$.

Step 4: information. The probe outcomes are i.i.d. given θ, so by Wald's identity the divergence of their law up to σ' under θ' from that under θ'' is $\mathbb{E}_{\theta'}[\sigma']\,\mathrm{kl}_{\theta'}(b_p) \leq n\,\mathrm{kl}_{\theta'}(b_p)$. The Bretagnolle–Huber inequality gives $P_{\theta'}(\text{declare } \theta'') + P_{\theta''}(\text{declare } \theta') \geq \frac{1}{2} e^{-n\,\mathrm{kl}_{\theta'}(b_p)}$, and with Step 3,

$$\frac{2n}{K} + \frac{8(\mathrm{Reg}_{\theta'} + \mathrm{Reg}_{\theta''})}{DK} \;\geq\; \tfrac{1}{2}\, e^{-n\,\mathrm{kl}_{\theta'}(b_p)}.$$

Step 5: two cases. Let $a := 1/(2\,\mathrm{kl}_{\theta'}(b_p))$. If $n \geq a \log K$, then

$$\mathrm{Reg} \;\geq\; \pi_{\theta'}\mathrm{Reg}_{\theta'} \;\geq\; \pi_{\theta'}\, r_{\theta'}(b_p)\, a \log K \;=\; c_\rho \log K.$$

If $n < a \log K$, then $e^{-n\,\mathrm{kl}} > K^{-1/2}$ and the display gives

$$\mathrm{Reg} \;\geq\; \min(\pi_H, \pi_L)\,(\mathrm{Reg}_{\theta'} + \mathrm{Reg}_{\theta''}) \;\geq\; \min(\pi_H, \pi_L)\Big(\frac{D}{16}\sqrt{K} - \frac{D}{4}\,a \log K\Big),$$

which exceeds $c_\rho \log K$ for K large. □

Remark 3 (Why the decomposition pays $\log K$ and the full bandit does not). Every auction is informative ($q_H(b) \neq q_L(b)$ for $b > 0$), so a team that keeps bidding near the state-contingent optimum learns the state as a by-product: its misclassification probability decays geometrically and its total loss is $O(1)$. The decomposition must commit after a separate probing phase, whose length has to grow like $\log K$ for the commitment error to cost $O(1)$ over the remaining $K - \tau$ auctions. Propositions 3(b) and 4 pin the decomposition's cost to order $\log K$, and Table 3 shows it growing by a roughly constant amount per doubling of K while the full bandit's stays flat. The lower bound needs only two features of the decomposition: probes at a bid that does not track the posterior, and an execution phase that stops learning. A standard refinement (exact error exponents instead of the Bretagnolle–Huber bound) suggests the slope $c_\rho^{\mathrm{p}}/\log(1/\beta)$, where $c_\rho^{\mathrm{p}} := \pi_H r_H(b^I) + \pi_L r_L(b^I)$ is the expected Lagrangian cost of one probe and $\log(1/\beta)$ is the Chernoff information of the probe outcome (the two outcome laws at b^I are mirror images, $q_H(b^I) = 1 - q_L(b^I)$); Table 4 compares it with the data.

Remark 4 (The budget rule matters more than the decomposition). With the paper's budget rule the decomposition's per-period premium converges instead to $\Psi(\rho)$ (Proposition 1; $\Psi = \max\{\mathrm{VPI}, 0\}$ at the baseline), so its learning premium relative to the full bandit tends to $\Psi(\rho)/\Lambda(\rho)$ (at $\rho = 10$: 0.43, 0.54, 0.66 for $r = 8, 16, 32$). At large budgets $\Psi(\rho) = \mathrm{VPI}(\rho) \approx \mathrm{Var}(\theta)/\rho^2$ (Proposition 2(b)) while $\Lambda(\rho) \approx \mathrm{Var}(\sqrt{\theta})/\rho$, so this ratio tends to zero: the gap grows with the budget and does not close as $K \to \infty$. The two rules differ in what information buys. Knowing the state, an ex-ante team can shift spend towards the state where the marginal dollar buys more (at large budgets it spends in proportion to $\sqrt{\theta}$), whereas under the paper's rule each state must spend its own budget. At small budgets the uninformed bid already spends across states almost as an informed ex-ante team would, which is why $\Lambda(\rho) = O(\rho^{3/2})$, and under the paper's rule information forces an equal split, which is why VPI < 0 there.

5.1 Numerical verification

We computed the full bandit (Bayes-optimal policy by dynamic programming over periods and posterior log-odds, with a Lagrange multiplier for the ex-ante budget) and the decomposition under the same rule for $r = 8, 16, 32$ ($\theta_H = 4, 8, 16$), $\pi_0 = \frac{1}{2}$ and $K = 50, 100, 200, 400$ (code: `bandit_check/run_dp_largeK.py`, `dp_convergence.py`, `region_map.py`). The asymptotics of (d) hold at the baseline: $\Lambda(\rho)/\rho$ falls from 0.0026 to 0.0002 as ρ falls from 10^{-2} to 10^{-4}, and $\rho\Lambda(\rho) = 1.123$ at $\rho = 10^4$ (theory: $\mathrm{Var}(\sqrt{\theta}) = 1.125$).

The ratio of learning premia rises with K at every budget (at $\rho = 5$: $0.08, 0.25, 0.45, 0.63$ for $r = 8$; $0.49, 0.65, 0.77, 0.86$ for $r = 16$; $0.75, 0.84, 0.90, 0.94$ for $r = 32$), and the budgets with ratio at least 0.8 widen on both sides, as (d) predicts: for $r = 16$ they are $\rho \in [0.95, 1.27]$, $[0.78, 2.08]$, $[0.64, 3.75]$ and $[0.53, 9.07]$ for $K = 50, 100, 200, 400$. The allocation shifts agree in direction on 84–93% of total budgets for $r \geq 16$, and on 28% ($K = 50$) to 88% ($K = 400$) for $r = 8$. The slowest object is the total budget at which learning switches from skewing to balancing the split, which depends on the junior team's small budget, where (d) makes the bound weakest: the decomposition's switch point moves towards the limit only for $r = 32$ ($t = 2.26, 2.39, 2.95, 3.42$ against 4.09), starts to move for $r = 16$ at $K = 400$ ($1.60 \to 1.92$ against 3.01), and has not moved

Table 3: Total learning cost $K\mathcal{L}(\rho) - V_K(\rho K)$ in impressions: decomposition / full bandit (ex-ante rule).

r	ρ	$K = 50$	$K = 100$	$K = 200$	$K = 400$
8	2.5	0.75 / 0.13	1.11 / 0.13	1.52 / 0.13	1.94 / 0.13
8	10	0.88 / 0.22	1.64 / 0.23	2.62 / 0.23	3.76 / 0.23
16	2.5	0.53 / 0.10	0.69 / 0.10	0.84 / 0.10	1.00 / 0.10
16	10	1.16 / 0.20	1.67 / 0.20	2.22 / 0.20	2.79 / 0.20
32	2.5	0.31 / 0.08	0.38 / 0.08	0.44 / 0.08	0.51 / 0.08
32	10	0.92 / 0.16	1.20 / 0.16	1.49 / 0.16	1.79 / 0.17

for $r = 8$ (≈ 1.07 against 2.32), while the full bandit's is within 0.11 of the limit already at $K = 50$. Between the two switch points the full bandit's shift is small, 0.25–0.54% of the total budget at $K = 100$.

Table 4: Growth of the decomposition's learning cost per unit of $\log K$: the proved lower bound c_ρ (Proposition 4, $b_p = b^I$), the observed slope between $K = 50$ and $K = 400$ (Table 3), the heuristic slope $c_\rho^{\mathrm{p}}/\log(1/\beta)$, and the slope of the proved upper bound $(1 + b^I/\rho)/\log(1/\beta)$.

r	ρ	lower bound c_ρ	observed	heuristic	upper bound
8	2.5	0.078	0.572	0.728	12.1
8	10	0.212	1.385	2.178	8.82
16	2.5	0.033	0.226	0.259	8.07
16	10	0.108	0.784	0.983	5.38
32	2.5	0.015	0.096	0.108	6.34
32	10	0.059	0.418	0.476	3.82

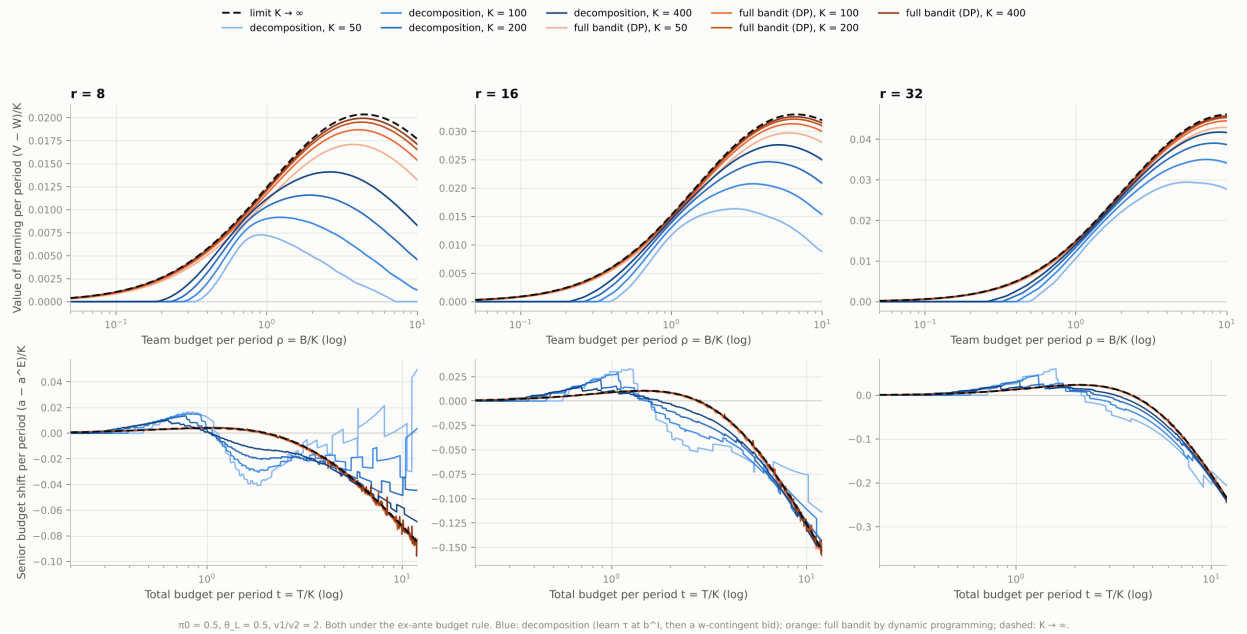


Figure 3: Ex-ante budget rule, $\pi_0 = \frac{1}{2}$. Top: per-period value of learning; bottom: per-period shift of the senior budget ($v_1/v_2 = 2$). Blue: decomposition, $K = 50, 100, 200, 400$ (light to dark); orange: full bandit; dashed: the limit $K \to \infty$ of Proposition 3.